Department of Economics

MSc Extended Essay Cover Sheet

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COVID-19 and Childcare: Heterogeneous impacts on hours worked

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May 31, 2022

Abstract

This paper uses the UK Labour Force Survey Data from 2018 to 2021 to investigate heterogeneous impacts of the COVID-19 crisis on total hours worked per week of people with and without children, by gender. Using a difference-in-differences model, I find evidence of a negative effect of having a child on total hours worked for both females and males, and a smaller positive effect of having a child during COVID-19 on total hours worked for both females and males. These effects are larger in magnitude for females. An extended model controlling for the age of the child finds that these effects are strongest for parents of children below the age of 2.

Keywords: gender inequality, COVID-19, childcare, total hours worked

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1. Introduction

As gender roles converge, gender equality has become an increasingly pertinent topic of discussion in recent years. As a result, labour outcomes such as the gender wage gap (GWG) have been extensively researched. In particular, some research has shown that the gender gap in hours worked can explain a substantial amount of the GWG if we consider average hours in an occupation (Denning et al, 2021; Cortes and Pan, 2019). Owing to social norms, women around the world take on the majority of unpaid care work (Ferrant et al, 2014) and often have to reduce their hours or stop working altogether upon becoming a mother in order to provide childcare, resulting in a motherhood wage penalty (Blau and Kahn, 2017; Paull, 2008). Indeed, there are significant child penalties on labour attachment for women and several research studies in the UK have shown that the GWG widened substantially in child-rearing years (Joshi et al, 2020; Viitanen, 2014).

The onset of COVID-19 poses new challenges for gender equality. It is a very peculiar shock – both economically and socially – that differs from previous recessions, where procyclical sectors such as construction and manufacturing are usually the hardest hit (Hupkau and Petrongolo, 2020). In the COVID-19 crisis, each sector is hit differently depending on whether it is considered essential and whether work can be carried out remotely by working from home (WFH). As governments imposed social distancing measures such as lockdowns, the way we go to work and live our lives was fundamentally changed. Lockdowns imposed school closures, which increased the childcare burden of parents while removing the possibility of outsourcing childcare. Given this development and established social norms, it is perhaps unsurprising that recent literature shows that on average, women provided a larger share of increased childcare needs (Hupkau and Petrongolo, 2020). However, lockdowns also created more opportunities for some mothers to WFH, which has been shown to reduce the gender gap in hours worked among parents (Arntz et al, 2019). In the US, telecommuting mothers

maintained paid work more than those who could not (Lyttelton et al, 2021). Given that the lack of flexible work arrangements had been identified as a big driver of the GWG in at least some industries like finance (Bertrand et al, 2010), the increase in flexible work arrangements may prove to be beneficial for gender equality. Yet, considering regional differences, it is unclear what the final impact on wages and hours worked may be in the UK. Given its peculiarities, COVID-19 can thus be seen as a unique opportunity – an exogenous shock – that can be used to investigate the relationship between having children and work hours. This is also a meaningful subject of study as evaluating whether impacts of COVID-19 are heterogenous for different groups can inform policymaking in order to improve gender equality under the changes brought about by COVID.

To provide further context of the development of COVID-19 and its accompanying social-distancing measures in the UK, the rough timeline is as follows. The UK went through multiple lockdowns, with the first beginning on 23 March 2020. This was eased in May 2020, but put in place again in October 2020 and then January 2021. Most restrictions were lifted by July 2021 as the situation stabilised before the Omicron wave hit in mid-December, 2021. The UK government also introduced the Coronavirus Job Retention Scheme on 20 March 2020 ("furlough"), where many employees worked zero hours but remained "employed".

As we begin to step back into the "new normal", this paper explores how parents are differentially impacted by COVID-19 compared to people with no children in the 2 years since the onset of COVID-19. I will compare the effects of COVID-19 on hours worked between those with or without children by gender. While the knock-on effects of COVID-19 will likely persist for several years, these preliminary findings may be indicative of potential future trajectories.

The paper proceeds as follows. Section 2 reviews the literature on gender inequality in the labour market and recent literature on COVID-19. Section 3 describes the data used and

discusses its summary statistics. Section 4 describes the empirical strategy used to identify the hypothesised heterogeneous effects of COVID-19 depending on whether one has children. Section 5 presents and analyses the main findings. Section 6 conducts some robustness checks and explains the limitations of this study, and Section 7 concludes.

2. Literature Review

There have been many efforts in recent decades to characterise and explain gender inequality in the labour market. Blau and Kahn (2017) find that gender differences in occupations and industries, as well as differences in gender roles remain important, while Magnusson (2016) finds significant GWG even in the same profession and controlling for specializations. This paper aims to build on the existing literature that studies various industries and argue that the gender gap in hours and preferences for job flexibility are important to understanding the GWG (Denning et al, 2021; Bertrand, Goldin and Katz, 2010; Cortes and Pan, 2019; Goldin, 2014). One major constraint on women's working hours stems from the event of having a child, which typically imposes significant childcare burdens on females and results in significant long-run penalties of close to 20% in female earnings after the birth of a child in both extensive and intensive margins (Kleven et al, 2019).

This makes COVID-19 a particular interesting subject of study as explained earlier. Despite its recency, there have been many studies documenting the impacts of COVID-19 in real time, with mixed results. For instance, Hupkau and Petrongolo (2020) find that labour market outcomes of men and women in the UK are roughly equally affected at the extensive margin, with women suffering smaller losses at the intensive margin. On the other hand, Blundell et al (2020) find workers in sector shutdowns are disproportionately female, young and low-paid, with an uneven distribution in the ability to continue working from home. Similarly, Andrew et al (2020a) find that in households with dependent children, mothers are

more likely than fathers to be out of work or furloughed during the crisis. In the US, Collins et al (2020) also found an increase in the gender gap in work hours, as mothers with young children reduced their work hours four to five times more than fathers. In contrast, Couch et al (2021) find that women had a greater likelihood to telework, higher education levels, and a less-impacted occupational distribution, which all contributed to lessening negative impacts relative to men.

In terms of the childcare burden, women provided a larger share of increased childcare needs on average, although in an important share of households fathers also became the primary childcare providers (Hupkau and Petrongolo, 2020). Similar findings are presented by Sevilla and Smith (2020), who find that in the UK, women working from home have undertaken more childcare than men during the pandemic, regardless of their employment status. Blundell et al (2020) also highlights that the economic shock associated with COVID-19 will interact with many pre-existing inequalities along dimensions such as gender, ethnicity, age and geography, with interrelated impacts across multiple related aspects in life such as health, jobs, and family. This highlights the complexity of impacts that we are faced with and possibly heterogeneous impacts on each individual.

There is also hope that an increase in the amount of childcare by men could kick off a change in social norms. For instance, in Australia, Craig and Churchill (2020) found that while absolute hours of care time for mothers increased more than fathers, the relative increase in men's care time resulted in narrowed gender gaps.

Recent research does not provide conclusive evidence of the impacts of COVID-19. Thus, this paper hopes to contribute to the literature by examining the effect of the COVID-19 crisis using the latest data in the context of UK and check for consistencies with existing studies. While many studies focus on gender wage gap, I focus directly on the hours worked and its

difference between people with and without children in order to identify heterogeneous impacts on these groups.

3. Data

The datasets used in this paper come from the UK Labour Force Survey (LFS). The LFS is a rotating panel survey conducted every quarter. Respondents are followed for five consecutive quarters and a fifth of respondents are replaced in each wave. Using the 2 quarter longitudinal datasets spanning the first quarter (Q1) of 2018 to the fourth quarter of 2021[1], I constructed an unbalanced panel dataset spanning 2018 to 2021. The sample size for each quarter is large and ranges from 21,550 in 2020Q2 to 28,897 in 2021Q2. The analysis in this paper is done using the provided longitudinal weights from when each respondent first entered the survey. Table 1[2] shows the list of main variables and their definitions in the dataset. Observations with missing values are dropped via listwise deletion.

I have restricted the sample to the population of interest, which includes respondents between the ages of 16 and 66[3], and excludes those who are between 16-18 in full time education and never married. The dependent variables of interest are total actual hours worked per week and the logarithm of net weekly wage. The summary statistics of the final sample used are shown in Table 2, showing results by gender and whether respondents have children. Employment rates are higher for males than for females, and higher for the group of respondents with children than the group of respondents without. The share of females working part time is significantly higher than males. Differences is shares of ethnicity and levels of

[1] The final unbalanced panel dataset was formed by merging 16 data files obtained from the UK Data Service.

[2] Please refer to all Tables and Figures in the Appendix.

[3] The State Pension age in the UK as of 2022 is 66.

education within each group are negligible, with females having slightly higher levels of education than males for the category "NQF Level ⩾4". Females also tend to have a greater average number of children across all groups, and a lower average net weekly wage than males across all groups.

As a brief overview of the results, Figure 1 shows that the average wage gap between females with and without children has narrowed, starting from 2020Q2[4] when the pandemic began in the UK. Figure 2 shows an increasing trend in the average wage gap between males with and without children throughout the 4 years, with no clear break in the trend in 2020Q2. The average wages of males with children is higher than that of males without children, while the average wages of females with children is lower than that of females without children. This corroborates existing literature on the fatherhood bonus and motherhood penalty (Hodges and Budig, 2010).

In Figure 3, total hours worked of females with and without children see a significant dip in 2020Q2, before increasing back to normal, with a smaller gap than before. In Figure 4, total hours worked of males with and without children face a similar dip in 2020Q2, however the gap in total hours worked for males, which is smaller than that for females to begin with, seems to have remained constant.

Finally, I do standard Oaxaca Decompositions at different points in time to observe changes in the GWG using cross-sectional quarterly LFS datasets.

$$w_m = \beta_m x_m + \epsilon_m$$

$$w_f = \beta_f x_f + \epsilon_f$$

$$\overline{w}_m - \overline{w}_f = \hat{\beta}\bar{x}_m - \hat{\beta}\bar{x}_f = (\hat{\beta}_m - \hat{\beta}_f)\bar{x}_m + (\bar{x}_m - \bar{x}_f)\hat{\beta}_f \tag{1}$$

[4] Time period 2020Q2 is indicated in the Figures 1 to 4 by a vertical dashed line.

Here, w refers to average logarithm of net weekly wages for each gender, x is a vector of average observed characteristics, and beta is the relevant coefficient. Observed characteristics here include age, age squared, highest level of qualification obtained, industry of main job, and whether one is working full time or part time.

The results of the Oaxaca Decomposition are presented in Table 3. While there seems to be a downward trend in the GWG from 2018 to 2019, the trend seems to have stagnated from 2020 to 2021.

One limitation of this dataset is that it is quarterly and cannot provide more granular data which would allow for more precise estimation. In addition, the collection of data for the LFS has been affected by COVID-19 – in March 2020, initial face-to-face interviews had to be changed to telephone-based interviews. This impacted both the level of response and the non-response bias of the survey, but has been addressed to some extent with updates to the LFS weighting methodology. In addition, despite the dataset being longitudinal with hours data being collected up to 5 times for each individual, wage data is collected only once for most individuals and at most twice for a small fraction of individuals. As a result, I was not able to conduct a similar difference-in-differences analysis on wage data, and given time and word limitations, I could not introduce a completely new model for wages. Further limitations are discussed in Section 6.

4. Empirical Strategy

In order to identify the heterogeneous impacts of COVID-19 on hours worked between respondents with and without children by gender, I use the difference-in-differences methodology to control for individual fixed effects, similar to that used by Couch et al (2021). Here, both groups experience the same COVID-19 shock, but for reasons outlined above (with COVID-19 having particular implications on childcare), I hypothesise that respondents with

children will experience "treatment" in terms of heterogeneous effects from the same exogenous shock. Thus, the control group here includes respondents with no children, while the treatment group includes respondents with children. I assume that selection into treatment is random – that is, respondents with children have similar characteristics as those without. While previous research showed that voluntarily childless adults have higher levels of education and greater attachment to the labour force (Bloom and Pebley, 1982), these will not affect the identification strategy here since these heterogeneities are largely time-invariant and will be absorbed by individual fixed effects.

The key underlying assumption required here is that there are common trends between the control and treatment groups – in other words, that there is no time-varying heterogeneity between the two groups (i.e. respondents with/without children). By visual inspection of Figures 3 and 4, the parallel trends assumption between respondents with and without children for total average hours worked before 2020 appears to hold for both males and females. That is, prior to COVID-19, employees with and without children experience similar trends in hours worked. Intuitively, there is reason to believe that the parallel trends assumptions hold since employees experience the similar macroeconomic shocks to work hours prior to COVID-19 regardless of whether they have children. This can be supported by the fact that there is "considerable persistence in the effect of children of work hours" on females (Paull, 2008), suggesting that due to this persistence, there is little time-varying heterogeneity on average between the two groups. For males, the event of having children has relatively little impact on work hours to begin with (Paull, 2008), so I would again not expect time-varying heterogeneity between the two groups. That being said, it is possible that there are policy shocks that affect respondents with and without children differentially. However, these can be partially ruled out on the basis that most of the government's efforts are being channelled into COVID-19 policy measures over the period of interest (2020-2021), and the effects of other policies will be

diminished. There have also been studies that show that a greater share of females work part time after having children (Paull, 2008), and this may result in differential responses to macroeconomic shocks. These issues should be kept in mind in the following analysis as they cannot be completely ruled out.

The baseline model is as follows:

$$y_{it} = \tau_t + \alpha_i + \beta CHILD_{it} \times COVID_t + \lambda X_{it} + \epsilon_{it} \mid gender = male/female \qquad (2)$$

With subscripts i denoting individual and t denoting time (by quarter). y_{it} is the outcome variable – total actual hours worked (includes zeros). Quarterly time fixed effects τ_t and individual fixed effects α_i are included as dummy variables. Individual fixed effects will absorb time invariant variables such as ethnicity, education level[5], industry of main job, and disability status. Time fixed effects assume that all observations are homogenously affected by a set of time-varying macroeconomic variables. X is individual time-varying control variable age. Separate regressions are run for each gender since labour market behaviour differs greatly between genders.

The COVID-19 variable is derived by setting a dummy variable to be 1 from 2020Q2 to 2021Q4. As observed in Figures 3 and 4, average total hours worked had not recovered to levels prior to COVID-19 as of 2021Q4. This is also seen in Table 4 with our regression results, where the value of coefficients on time fixed effects had not returned to pre-COVID levels, suggesting that their impact had not worn off, although government restrictions on COVID-19 had mostly been relaxed by July 2021 in the UK[6].

In the baseline model, level effects for COVID-19 are not included since they are derived from the time variable and will be completely absorbed by time fixed effects, which

[5] Education levels are largely time invariant for adults who are not in full time education.

[6] While there were still some measures such as compulsory face masks in place, legal limits on social contact have been removed; schools and the last closed sectors of the economy reopened, including nightclubs.

capture time shocks that are common across all cross-sectional units. Level effects for CHILD are not included initially since they too may be absorbed by individual fixed effects if time invariant. Furthermore, the decision to have children – if affected by COVID-19 – could be endogenous. However, recent data shows no evidence of any large, sustained change in fertility and only a short-term decline in births in the UK that was already the trend before COVID-19 (Berrington et al, 2022), lowering the likelihood of endogeneity from CHILD level effects. A quick robustness check by dropping observations with children below the age of 2 shows results with the same sign but a lower magnitude and level of significance. Thus, in further specifications, I include level effects for CHILD to investigate if it will be absorbed by individual fixed effects, since it is a time varying variable. This specification is modelled as follows:

$$y_{it} = \tau_t + \alpha_i + \gamma CHILD_{it} + \beta CHILD_{it} \times COVID_t + \lambda X_{it} + \epsilon_{it} \mid gender = male/female \qquad (3)$$

I have considered variables that indicate whether the employee is working from home, and whether the employee is working full time or part time. However, these variables have not been included in the regression since they are potentially endogenous and may affect both the dependent variable (hours worked) as well as independent variables (CHILD, and CHILD*COVID). For instance, females with childcare duties are more likely to work part time, thus affecting hours worked, while an indicator of whether one works from home would be correlated with the dummy variable COVID. Thus, these 2 variables would be bad controls as they could be dependent variables of CHILD and CHILD*COVID. I have not included age squared in this model since the dependent variable here is total hours worked, and unlike the case for wages, it is not likely for there to be a non-linear relationship between age and hours since there is an upper limit to the amount of hours that can be worked.

Finally, I extend the model to separate effects of children of different age bands on hours worked since it is likely that children in different age bands require different levels of

care. The age bands are as follows: below 2 (infant), 2-4 (preschool), 5-9 (primary), 10-16 (secondary), 17-19 (further education; sixth form). I use dummy variables to indicate whether each respondent has a child in relevant age band. This extension is modelled as follows, with 2 specifications – one including CHILD level effects, and the other not.

$$y_{it} = \tau_t + \alpha_i + \sum_{j=1}^{5} \gamma_j CHILD_{ijt} + \sum_{j=1}^{5} \beta_j CHILD_{ijt} \times COVID_t + \lambda X_{it} + \epsilon_{it} \qquad (4)$$

where j corresponds to each of the 5 age bands specified above.

5. Results

The main results on total actual hours worked can be found in Table 4, estimated using OLS. Specifications 1 to 3 are for females, while specifications 4-6 are for males. Specification 1 and 4 omits the time-varying control (age) and CHILD level effects. Specification 2 and 5 includes age but not CHILD level effects, while specification 3 and 6 includes both age and CHILD level effects.

Impact on females:

In regressions 1-2, I find that the overall average effect of the interaction term CHILD*COVID (i.e. having a child and being in COVID time periods) is to decrease total actual hours worked by 0.561 hours, significant at the 5% level. However, when I include level effects for CHILD (regression 3), the sign flips and the overall average effect of CHILD*COVID is to increase total actual hours worked slightly by 0.909 hours, significant at the 1% level. At the same time, the level effects of having a child is to decrease total actual hours worked by 2.220 hours. Given that there is no evidence of large changes in fertility as mentioned above, this discrepancy may be due to the fact that omitting the level effects for CHILD resulted in an omitted variable bias, and it is only by controlling for these level effects that the effects of having a child can be separated from the heterogeneous effects of COVID-19 on females with children. In addition, the control variable AGE has no significant impact

on the total actual hours worked, and dropping the AGE control has little impact on the values of other coefficients.

The finding of a positive interaction effect of CHILD*COVID is surprising, but consistent with the results in Figure 3, which shows that the gap in total actual hours worked between females with and without children have narrowed since the onset of COVID-19. This suggests that while having a child has a negative impact on the hours worked for females, COVID-19 has actually alleviated the situation for those with children *relative* to those without children. One possible explanation for this positive effect could be that COVID-19 and the switch to WFH had given those with children more flexibility to fulfil childcare duties. The importance of flexibility to the GWG has been discussed in past literature (Denning et al, 2021, Goldin and Katz, 2016). This flexibility may often be more useful than a reduction of hours. For instance, while before, parents had to fetch their children from school at fixed times, they can now fulfil their childcare duties (e.g. watching over their children) while working at the same time, or shift their work to later hours of the day after their children have gone to bed. As such, they are able to accomplish more hours of work in a day despite taking on more childcare duties. For those with no children, this extra flexibility is simply a bonus but not crucial. Thus in relative terms, there is a positive effect for those with children. Another way to reconcile this finding with research that shows females taking on more childcare burdens is that they are taking on more childcare burdens and doing more work at the same time, which may be the case especially with less clear boundaries on when to work and not work when telecommuting.

Finally, there is a large increase in the magnitude of the time fixed effects in all 3 specifications from 2020Q2, and these are all significant at the 0.1% level. For instance, in specification 3, the fixed effect for 2020Q2 is -9.500, compared to -2.706 in the previous period. This shows that COVID-19 was indeed a sizeable shock to hours worked. These time fixed effects are slowly waning off but remains large in magnitude even by the end of 2021. This

may imply that while measures have been relaxed, the economy has not completely recovered and there remains persistent scarring effects from the pandemic and recession.

Impact on males:

From regressions 4-5 of Table 4, I find that the effect of CHILD*COVID on total actual hours is 0.157 and 0.159 respectively, and not significant. After including level effects of CHILD in regression 6, the overall average effect of CHILD*COVID is to increase total actual hours worked by 0.817 hours, significant at the 5% level, while CHILD level effects serve to decrease total actual hours worked by 0.989 hours, significant at the 0.1% level. This result is consistent with that of females in that the effect of CHILD*COVID is positive with a greater level of significance after adding in CHILD level effects. From Figure 4, there is no clear narrowing of total actual hours worked between males with and without children since the beginning of COVID. This is in line with the smaller coefficient on CHILD*COVID here compared to that for females. Nevertheless, there is a small positive effect which suggests that male caregivers also benefit to some extent from the increased flexibility to fulfil childcare duties.

Unlike the case for females, age results in a 0.410 hour decrease in total actual hours worked for males, and is significant at the 0.1% level. That is, as age increases, total actual hours worked decreases. One possible explanation for this is that there is a greater share of men in senior roles which typically require lower hours of work, and as men age and get promoted into senior roles, their hours worked decreases. For occupations requiring manual labour, it is also likely that hours worked decrease given that physical abilities decrease as age increases.

Finally, I observe that there is a large increase in magnitude of the coefficients of time fixed effects for males in all 3 specifications from 2020Q2. For instance, in specification 6, the time fixed effect for 2020Q2 is -10.45, compared to -2.650 in the previous period. In addition, time fixed effects for males are larger than that of females for the periods from 2020Q2, which

suggests that males on average have sustained a larger decrease in actual hours of worked due to COVID-19.

Next, I would like to compare the regression results between males and females. Firstly, both CHILD and CHILD*COVID effects are larger in magnitude for females than males. This is in line with the literature and our knowledge of social norms where mothers are the primary caretakers of their children. Another interesting point is that age is not a significant independent variable for females, but is very significant for males. Assuming that the explanation that those in higher positions work less is true in this case, this is also suggestive that higher positions are disproportionately taken by men. Finally, comparing time fixed effects between males and females (i.e. regressions 3 and 6), males seem to have recovered from the recession better than females, as time fixed effects for males are smaller in magnitude that females starting from 2020Q3.

Extended model

Next, I look at results on the extended model from equation 4, where total actual hours worked is regressed on categorical dummy variables of the age band of the child. Results are presented in Table 5, with specifications 1 and 2 for females, and specifications 3 and 4 for males. Specification 1 and 3 omits the CHILD level effects, while specification 2 and 4 includes CHILD level effects. Comparing Tables 4 and 5, time fixed effects are similar for regressions in both models. Thus, these time fixed effects are robust to the addition of controls for the age of the CHILD.

Impact on females:

I find a negative effect on CHILD2*COVID in regression 1 and positive effect on CHILD2*COVID in regression 2 upon inclusion of CHILD level effects, similar to what is seen in Table 4. For females, the effect of the level effect of having a child under the age of 2 is extremely large at -7.319, and is significant at the 0.1% level, while the negative effect is

between 0 and -1.216 for children in other age bands. This suggests that the effect of having children on hours worked for females is most intense in the 2 years after birth, and decreases over time. This is in line with the hypothesis that levels of care required for children of different ages differ.

The coefficient on the interaction effect of having a child under the age 2 of in regression 2 is 3.404, positive and significant at the 0.1% level, while the rest of the coefficients on interaction effects between COVID and different age bands are insignificant. This may be evidence for an explanation as follows: the COVID-19 crisis and the move to WFH had a large benefit for mothers with very young children under the age of 2. Where they would otherwise not be able to work at all given the extremely taxing demands of an infant, once everyone is working from home they can now put in more hours of work than before due to the increased flexibility which is particularly needed when children are young, and especially so infants who need constant care. Hupkau and Petrongolo (2020) find that 24% of all employed women work in jobs that allow WFH. This group of female employees who can WFH are the ones who can benefit most from the WFH policy, and while this is not the majority, due to the large magnitude of this effect on hours worked, it may be substantial enough to affect the overall average effect on hours worked (which includes effects on those who cannot WFH). Given this explanation, the interpretation in Table 4 may have to be reconsidered, since children of different age bands have different impacts on hours worked and this may mean that the average effect in table 4 is not indicative of a similar experience across the board.

Impact on males:

For males, the negative level effect of having a child under the age of 2 is -1.273, the largest compared to having children of other age bands and significant at the 1% level, but still much smaller compared to that for females. This is in line with research done by Kleven et al (2019), who show that hours worked of males remain stable after the event of a child birth.

Interaction effects for males are also insignificant except for that for children between the ages of 2 to 4, with a positive impact of 1.204 at the 5% level of significance. The fact that interaction effects for males are largely insignificant further supports the explanation given above for females. As fathers' hours are not as strongly affected by the birth of a child even without flexible WFH measures, the extra flexibility offered by COVID-19 is not as pertinent for them. There is also a similar negative effect of age on hours worked as the baseline model in Table 4 at -0.412 hours, significant at the 0.1% level.

6. Robustness Checks and Limitations

In order to check for the robustness of the main results, I vary the COVID variable by setting COVID=0 for the last 2 quarters of 2021, given that most restrictions have been relaxed in the UK by that time, and cases had somewhat stabilised before the spread of the Omicron variant. From Table 6, the main results from regressions 3 and 6 are found to be robust to varying COVID time periods, as coefficients on CHILD*COVID are similar to those in Table 4. The same cannot be said for regressions 1 and 2, where the CHILD*COVID coefficients become closer to zero and insignificant. In regressions 4 and 5, CHILD*COVID coefficients increase significantly compared to the same regressions in Table 5. The robustness of regressions 3 and 6 lends support to their validity, and is supported by the earlier analysis that regressions 1, 2, 4, 5 suffer from omitted variable bias.

There are also several limitations in this essay. Given that I only have quarterly, rather than more granular monthly data, I could not control for the severity of COVID-19 and changes in social-distancing policies. For instance, an important social-distancing policy is school closures in the UK. These spanned from the last week of March to the second week of June in 2020, and from January to the first week of March in 2021. Together with regional differences in a tiered system of policies as well as phased returns to school, these could be important in

explaining differences in hours worked, but could not be accurately captured by the model given that time periods were measure by quarters. Furthermore, due to the nature of studying a natural experiment, the data is noisy and not all variables can be controlled for. For instance, given the lack of data, it is difficult to untangle the effects of COVID-19 (for instance, if respondents caught COVID-19 and had to cut back on hours worked) and the measures put in place to target COVID-19, since these measures are endogenous to the progression of the pandemic. Finally, the true impacts of COVID-19 may differ for every individual and may affect people in different ways that cannot be fully captured by the model.

7. Conclusion

The COVID-19 pandemic has brought new challenges and possibly fundamental changes to the way people work. In this process, it is important for us to untangle heterogeneous impacts as best we can. This paper hopes to provide such a discussion, and examines the heterogeneity of impacts on people with and without children. The analysis is done on data from the LFS for 2018Q1 to 2021Q4, using a difference-in-differences methodology.

I have found evidence that during the pandemic, there is on average a positive impact of having a child on hours for both males and females. For females, there is an increase of 0.909 hours per week; for males, there is an increase of 0.817 hours per week. Note that overall, there are still strong negative impacts of having a child on total hours worked (-2.220 hours for females and -0.989 hours for males), and this is not outweighed by the small positive impacts of having a child during COVID-19. While initially surprising, this finding is supported by the narrowing hours and wage gaps between females with and without children, and suggests that the flexibility that COVID-19 provides is particularly beneficial for fulfilling childcare duties, especially for females. When I run an extended regression controlling for the age of the child, I find the strongest positive impact on those who have children below the age of 2 for females,

with an increase of 3.404 hours per week, while impacts of having children in other age bands and for males are insignificant. This is in line with expectations of infants requiring greater amounts of childcare and thus imposing the largest impact on hours worked for parents.

These estimates are robust to changes in the COVID-19 dummy variable. Nevertheless, one should keep in mind that there are limitations to these results due to insufficient granularity and noise in the data. On the whole, increased flexibility in working arrangements can be a game changer in reducing the hours gap even after the other impacts of the pandemic has worn off.

In conclusion, the results in this paper suggest that there are heterogeneous impacts of COVID-19 on people with and without children, but these impacts are stronger for females with children below the age of 2. While it is too still early to see longer term effects, given the drastic change in the way we work, there is potential for social norms to change in future, especially with more men having taken up the role as the primary childcare provider. These persistent effects of COVID-19 can be further investigated in future research.

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9. Appendices

Figure 1: Logarithm of Average Wages for Females by whether one has children, 2018Q1-2021Q4

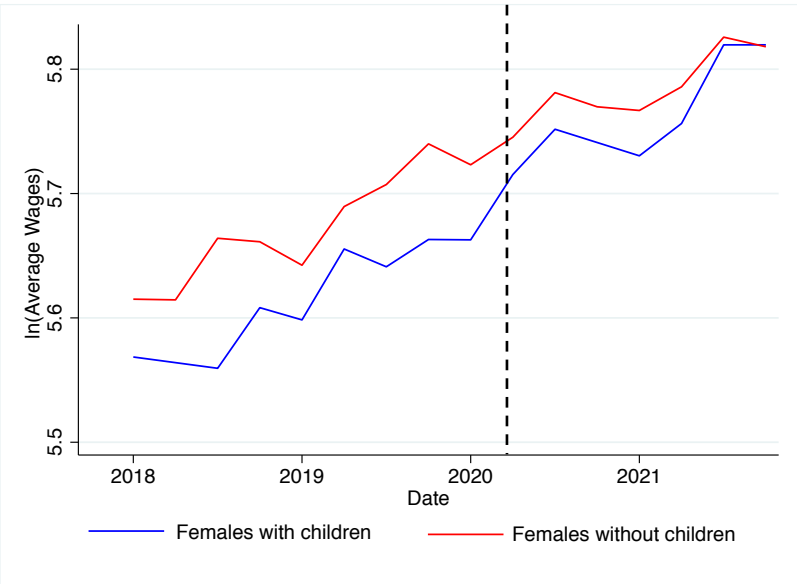

Figure 2: Logarithm of Average Wages for Males by whether one has children, 2018Q1-2021Q4

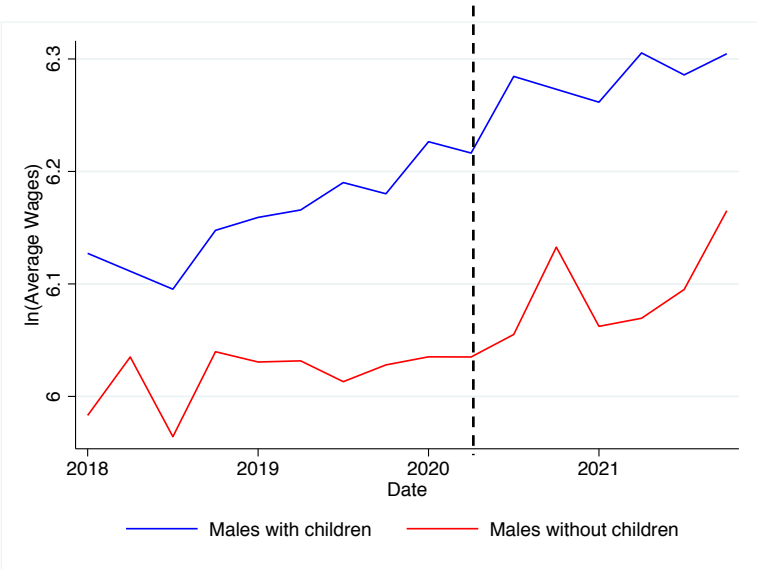

Figure 3: Total average hours worked for Females by whether one has children, 2018Q1-2021Q4

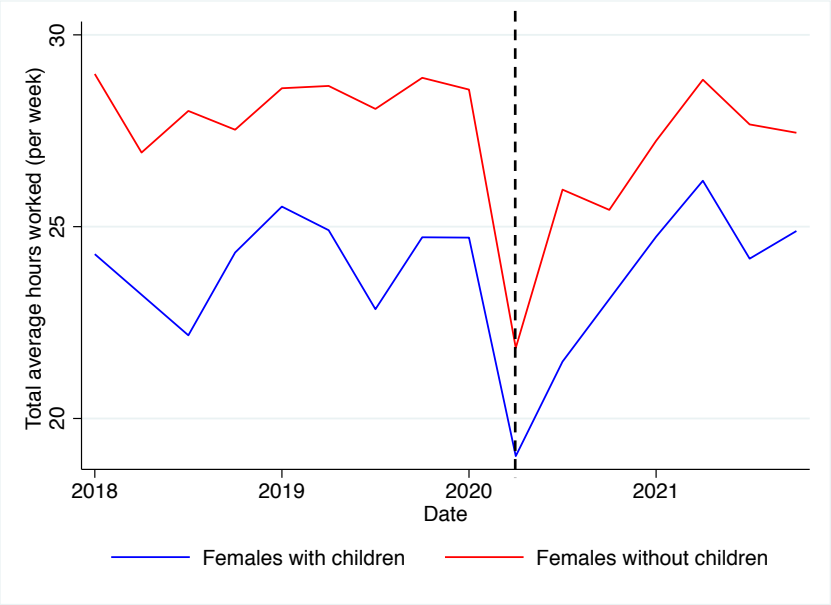

Figure 4: Total average hours worked for Males by whether one has children, 2018Q1-2021Q4

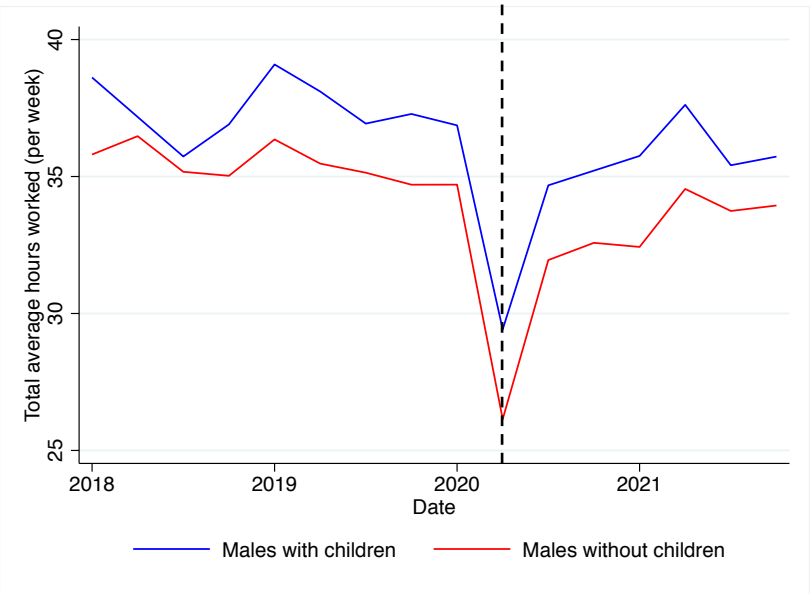

Table 1: List of Main Variables

Variable	**Description**
CHILD	Binary variable, =1 if number of dependent children in family aged under 19 > 0, =0 otherwise.
CHILD2	Binary variable, =1 if number of dependent children in family aged between under 2 >0, =0 otherwise.
CHILD4	Binary variable, =1 if number of dependent children in family aged between 2 to 4 >0, =0 otherwise.
CHILD9	Binary variable, =1 if number of dependent children in family aged between 5 to 9 >0, =0 otherwise.
CHILD16	Binary variable, =1 if number of dependent children in family aged between 10 to 15 >0, =0 otherwise.
CHILD19	Binary variable, =1 if number of dependent children in family aged between 16 to 18 >0, =0 otherwise.
COVID	Binary variable, =1 for time periods from 2020Q2 to 2021Q4, =0 otherwise.
CHILD*COVID	Interaction variable between CHILD and COVID.
AGE	Integer value, age of respondent at date of survey.
Hours Worked	Real number ≥ 0, total actual hours worked in main job in reference week
Education	Categorical variable describing level of highest education qualification attained based on the National Qualification Framework (NQF).
Ethnicity	Categorical variable, ethnicity in categories: White, Mixed, Indian, Pakistani, Bangladeshi, Chinese, Other Asian, Black/Black British, Other
Full time/ Part time	Categorical variable, in categories: Full time, Part time.
Employed	Categorical variable, in categories: In employment, ILO unemployed, Inactive, Under 16

Table 2: Summary Statistics of LFS Panel Dataset, 2018Q1 to 2021Q4

	All				With Children				Without Children			
	Male		Female		Male		Female		Male		Female	
	Share	Std. Dev.	Share	Std. Dev.	Share	Std. Dev.	Share	Std. Dev.	Share	Std. Dev.	Share	Std. Dev.
Employed	0.781	0.414	0.705	0.456	0.848	0.359	0.740	0.439	0.736	0.441	0.678	0.467
Full time (conditional on working)	0.885	0.319	0.593	0.491	0.907	0.291	0.532	0.499	0.867	0.339	0.637	0.481
Part time	0.114	0.319	0.407	0.491	0.093	0.291	0.468	0.499	0.133	0.339	0.363	0.481
Education												
⩾NQF Lvl 4	0.425	0.494	0.476	0.499	0.450	0.497	0.503	0.500	0.406	0.491	0.455	0.498
NQF Lvl 3	0.166	0.372	0.155	0.362	0.165	0.371	0.164	0.370	0.167	0.373	0.150	0.357
Trade Apprenticeships	0.061	0.240	0.013	0.112	0.052	0.221	0.012	0.109	0.068	0.251	0.013	0.115
NQF Lvl 2	0.124	0.330	0.152	0.359	0.129	0.335	0.146	0.353	0.123	0.328	0.158	0.364
<NQF Lvl 2	0.097	0.295	0.097	0.297	0.090	0.286	0.082	0.275	0.101	0.301	0.108	0.311
Other Quals	0.056	0.230	0.042	0.202	0.058	0.233	0.043	0.202	0.056	0.229	0.043	0.202
No Quals	0.071	0.256	0.064	0.245	0.057	0.231	0.050	0.219	0.079	0.270	0.074	0.262
Ethnicity												
White	0.914	0.280	0.904	0.294	0.887	0.317	0.876	0.330	0.930	0.254	0.923	0.266
Mixed	0.009	0.094	0.010	0.100	0.010	0.098	0.011	0.104	0.009	0.092	0.009	0.096
Indian	0.020	0.140	0.020	0.139	0.027	0.162	0.252	0.157	0.016	0.125	0.016	0.126
Pakistani	0.011	0.106	0.011	0.104	0.018	0.134	0.171	0.130	0.007	0.084	0.007	0.083
Bangladeshi	0.003	0.063	0.004	0.061	0.007	0.082	0.006	0.077	0.002	0.047	0.002	0.046
Chinese	0.004	0.065	0.006	0.778	0.005	0.071	0.007	0.085	0.003	0.063	0.005	0.073
Other Asian	0.008	0.089	0.010	0.100	0.010	0.100	0.013	0.112	0.006	0.082	0.008	0.090
Black/Black British	0.018	0.133	0.023	0.150	0.022	0.146	0.029	0.168	0.016	0.124	0.019	0.136
Other	0.011	0.104	0.012	0.110	0.014	0.118	0.016	0.125	0.009	0.094	0.010	0.099

Observations	234529	262038	109181	128566	170495	183567

	All				With Children				Without Children			
	Male		Female		Male		Female		Male		Female	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Dependents aged <19	0.596	0.961	0.649	0.983	1.763	0.820	1.752	0.824	N.A.	N.A.	N.A.	N.A.
Dependents aged <16	0.512	0.897	0.559	0.921	1.514	0.927	1.509	0.924	N.A.	N.A.	N.A.	N.A.
Dependents aged 10-15	0.209	0.532	0.229	0.549	0.618	0.764	0.617	0.757	N.A.	N.A.	N.A.	N.A.
Dependents aged 5-9	0.162	0.453	0.180	0.473	0.480	0.675	0.485	0.675	N.A.	N.A.	N.A.	N.A.
Dependents below 2-4	0.090	0.312	0.096	0.322	0.265	0.492	0.260	0.488	N.A.	N.A.	N.A.	N.A.
Dependents below 2	0.051	0.228	0.054	0.235	0.151	0.372	0.147	0.369	N.A.	N.A.	N.A.	N.A.
Age	45.94	13.64	46.20	13.38	43.18	12.09	42.06	11.53	47.60	14.37	48.97	13.92
Observations	234529		262038		109181		128566		170495		183567	
Net weekly wage	536.2	659.9	366.6	398.2	566.9	474.3	361.7	292.8	511.1	725.4	369.0	431.1
Observations	42361		48198		22554		25857		29114		32801	

Table 3: Oaxaca Decompositions, 2018Q1 – 2021Q4

	2018Q1	2018Q2	2018Q3	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Differential								
Male	6.006***	6.032***	6.025***	6.055***	6.047***	6.060***	6.063***	6.068***
	(0.00914)	(0.00896)	(0.00911)	(0.00896)	(0.00913)	(0.00900)	(0.00936)	(0.00945)
Female	5.585***	5.615***	5.619***	5.636***	5.619***	5.655***	5.676***	5.685***
	(0.00916)	(0.00916)	(0.00888)	(0.00935)	(0.00917)	(0.00882)	(0.00907)	(0.00930)
Difference	0.421***	0.417***	0.406***	0.418***	0.427***	0.405***	0.387***	0.383***
	(0.0129)	(0.0128)	(0.0127)	(0.0129)	(0.0129)	(0.0126)	(0.0130)	(0.0133)
Decomposition								
Endowments	0.168***	0.180***	0.184***	0.184***	0.206***	0.192***	0.175***	0.150***
	(0.00979)	(0.00966)	(0.00954)	(0.00996)	(0.00975)	(0.00944)	(0.00986)	(0.00972)
Coefficients	0.196***	0.205***	0.204***	0.209***	0.193***	0.159***	0.215***	0.138***
	(0.0132)	(0.0132)	(0.0132)	(0.0135)	(0.0135)	(0.0130)	(0.0134)	(0.0138)
Interaction	0.0571***	0.0314**	0.0178	0.0246*	0.0290**	0.0541***	-0.00285	0.0946***
	(0.0105)	(0.0105)	(0.0102)	(0.0109)	(0.0107)	(0.0104)	(0.0105)	(0.0112)
Observations	9822	9587	9366	9583	9523	9717	9358	9322

	2020Q1	2020Q2	2020Q3	2020Q4	2021Q1	2021Q2	2021Q3	2021Q4
Differential								
Male	6.075***	6.075***	6.133***	6.119***	6.147***	6.145***	6.162***	6.189***
	(0.0101)	(0.0112)	(0.00977)	(0.00958)	(0.00918)	(0.00950)	(0.00921)	(0.00951)
Female	5.671***	5.677***	5.765***	5.749***	5.761***	5.769***	5.805***	5.806***
	(0.0103)	(0.0109)	(0.00951)	(0.00943)	(0.00930)	(0.00941)	(0.00889)	(0.00971)
Difference	0.404***	0.398***	0.369***	0.370***	0.386***	0.376***	0.357***	0.383***
	(0.0144)	(0.0156)	(0.0136)	(0.0134)	(0.0131)	(0.0134)	(0.0128)	(0.0136)
Decomposition								
Endowments	0.201***	0.197***	0.180***	0.169***	0.188***	0.113***	0.151***	0.170***
	(0.0110)	(0.0118)	(0.0101)	(0.0103)	(0.0100)	(0.00945)	(0.00920)	(0.00997)
Coefficients	0.136***	0.109***	0.146***	0.230***	0.231***	0.243***	0.163***	0.160***
	(0.0152)	(0.0167)	(0.0144)	(0.0134)	(0.0133)	(0.0138)	(0.0130)	(0.0147)
Interaction	0.0671***	0.0916***	0.0431***	-0.0281**	-0.0328**	0.0195	0.0429***	0.0537***
	(0.0124)	(0.0138)	(0.0114)	(0.0101)	(0.0104)	(0.0102)	(0.00991)	(0.0117)
Observations	8000	7086	8543	9207	8843	8595	9441	8192

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.00$

Table 4: Impact of COVID-19 on Total Actual Hours Worked

	(1)	(2)	(3)	(4)	(5)	(6)
CHILD			-2.220***			-0.989***
			(0.216)			(0.229)
AGE		0.00799	0.00496		-0.410***	-0.410***
		(0.156)	(0.157)		(0.116)	(0.116)
CHILD=1* COVID=1	-0.561*	-0.561*	0.909**	0.157	0.159	0.817*
	(0.268)	(0.268)	(0.302)	(0.300)	(0.300)	(0.340)
2018Q2	-0.252	-0.254	-1.221***	-0.640**	-0.538**	-0.942***
	(0.171)	(0.176)	(0.206)	(0.198)	(0.201)	(0.225)
2018Q3	-1.915***	-1.919***	-1.950***	-1.759***	-1.548***	-1.561***
	(0.219)	(0.234)	(0.234)	(0.252)	(0.258)	(0.259)
2018Q4	-1.150***	-1.156***	-1.576***	-2.031***	-1.720***	-1.891***
	(0.244)	(0.270)	(0.276)	(0.286)	(0.299)	(0.302)
2019Q1	-0.192	-0.200	-0.837**	-0.881**	-0.473	-0.746*
	(0.266)	(0.306)	(0.317)	(0.298)	(0.322)	(0.329)
2019Q2	-1.272***	-1.282***	-1.739***	-2.150***	-1.636***	-1.833***
	(0.289)	(0.345)	(0.351)	(0.341)	(0.375)	(0.379)
2019Q3	-2.721***	-2.733***	-3.195***	-2.626***	-2.005***	-2.203***
	(0.320)	(0.391)	(0.397)	(0.372)	(0.416)	(0.419)
2019Q4	-2.017***	-2.031***	-2.492***	-2.811***	-2.091***	-2.288***
	(0.344)	(0.432)	(0.438)	(0.389)	(0.444)	(0.447)
2020Q1	-2.236***	-2.252***	-2.706***	-3.271***	-2.455***	-2.650***
	(0.366)	(0.473)	(0.479)	(0.416)	(0.480)	(0.483)
2020Q2	-8.402***	-8.420***	-9.500***	-10.90***	-9.978***	-10.45***
	(0.426)	(0.546)	(0.564)	(0.485)	(0.558)	(0.572)
2020Q3	-5.946***	-5.966***	-7.041***	-6.119***	-5.091***	-5.563***
	(0.440)	(0.582)	(0.599)	(0.496)	(0.580)	(0.593)
2020Q4	-5.775***	-5.797***	-7.174***	-6.835***	-5.700***	-6.305***
	(0.450)	(0.616)	(0.641)	(0.503)	(0.603)	(0.621)
2021Q1	-4.756***	-4.780***	-5.857***	-5.241***	-4.006***	-4.481***
	(0.478)	(0.662)	(0.678)	(0.531)	(0.640)	(0.652)
2021Q2	-3.741***	-3.767***	-4.854***	-3.800***	-2.461***	-2.940***
	(0.494)	(0.701)	(0.716)	(0.552)	(0.676)	(0.687)
2021Q3	-5.466***	-5.494***	-6.584***	-5.551***	-4.108***	-4.589***
	(0.512)	(0.743)	(0.758)	(0.573)	(0.710)	(0.721)
2021Q4	-5.584***	-5.615***	-6.842***	-5.861***	-4.318***	-4.860***
	(0.528)	(0.782)	(0.800)	(0.588)	(0.741)	(0.754)
Constant	30.33***	30.02***	31.44***	38.81***	54.43***	54.98***
	(0.286)	(6.017)	(6.047)	(0.326)	(4.414)	(4.416)
Observations	163049	163049	163049	148648	148648	148648
R^2	0.017	0.017	0.018	0.020	0.020	0.020

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Impact of COVID-19 on Total Actual Hours Worked, controlling for age of child

	(1)	(2)	(3)	(4)
CHILD2		-7.319***		-1.273**
		(0.431)		(0.413)
CHILD4		-0.672*		-0.985**
		(0.313)		(0.360)
CHILD9		-0.483		-0.614*
		(0.263)		(0.304)
CHILD16		-1.083***		-1.108**
		(0.309)		(0.364)
CHILD19		-1.216**		-0.653
		(0.452)		(0.543)
AGE	0.0104	-0.00558	-0.410***	-0.412***
	(0.156)	(0.158)	(0.116)	(0.116)
CHILD2=1*COVID=1	-2.109***	3.404***	-1.007	-0.0796
	(0.589)	(0.638)	(0.569)	(0.661)
CHILD4=1*COVID=1	0.164	0.133	0.574	1.204*
	(0.418)	(0.474)	(0.489)	(0.560)
CHILD9=1*COVID=1	-0.174	0.129	0.00998	0.368
	(0.359)	(0.400)	(0.427)	(0.476)
CHILD16=1*COVID=1	-0.242	0.537	0.157	0.915
	(0.388)	(0.442)	(0.447)	(0.519)
CHILD19=1*COVID=1	-0.410	0.502	0.274	0.711
	(0.508)	(0.634)	(0.634)	(0.783)
2018Q2	-0.255	-1.129***	-0.538**	-0.993***
	(0.176)	(0.201)	(0.201)	(0.223)
2018Q3	-1.920***	-1.965***	-1.548***	-1.570***
	(0.234)	(0.233)	(0.258)	(0.259)
2018Q4	-1.158***	-1.547***	-1.721***	-1.917***
	(0.270)	(0.274)	(0.299)	(0.302)
2019Q1	-0.203	-0.801*	-0.473	-0.791*
	(0.306)	(0.314)	(0.322)	(0.329)
2019Q2	-1.285***	-1.709***	-1.637***	-1.870***
	(0.345)	(0.348)	(0.375)	(0.378)
2019Q3	-2.735***	-3.158***	-2.006***	-2.240***
	(0.391)	(0.394)	(0.416)	(0.419)
2019Q4	-2.034***	-2.431***	-2.091***	-2.321***
	(0.432)	(0.435)	(0.444)	(0.447)
2020Q1	-2.262***	-2.646***	-2.457***	-2.688***
	(0.473)	(0.475)	(0.480)	(0.483)
2020Q2	-8.467***	-9.417***	-9.933***	-10.47***
	(0.544)	(0.557)	(0.557)	(0.569)
2020Q3	-6.015***	-6.949***	-5.047***	-5.579***
	(0.580)	(0.592)	(0.579)	(0.591)
2020Q4	-5.810***	-6.993***	-5.696***	-6.375***
	(0.616)	(0.635)	(0.603)	(0.619)
2021Q1	-4.834***	-5.769***	-3.969***	-4.504***
	(0.661)	(0.674)	(0.639)	(0.650)
2021Q2	-3.826***	-4.767***	-2.430***	-2.969***

	(0.700)	(0.712)	(0.675)	(0.685)
2021Q3	-5.552***	-6.494***	-4.073***	-4.614***
	(0.742)	(0.754)	(0.710)	(0.719)
2021Q4	-5.648***	-6.698***	-4.296***	-4.903***
	(0.782)	(0.797)	(0.741)	(0.753)
Constant	29.93***	31.72***	54.41***	55.12***
	(6.015)	(6.085)	(4.415)	(4.415)
Observations	163049	163049	148648	148648
R^2	0.017	0.022	0.020	0.021

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Robustness checks by varying COVID variable

	(1)	(2)	(3)	(4)	(5)	(6)
CHILD			-2.133***			-0.962***
			(0.204)			(0.216)
AGE		0.00775	0.00527		-0.411***	-0.410***
		(0.156)	(0.157)		(0.116)	(0.116)
CHILD=1* COVID=1	-0.000712	-0.000717	0.949***	0.587*	0.589*	1.017***
	(0.244)	(0.244)	(0.258)	(0.275)	(0.275)	(0.290)
2018Q2	-0.252	-0.254	-1.183***	-0.640**	-0.538**	-0.931***
	(0.171)	(0.176)	(0.203)	(0.198)	(0.201)	(0.222)
2018Q3	-1.915***	-1.919***	-1.949***	-1.759***	-1.548***	-1.561***
	(0.219)	(0.234)	(0.234)	(0.252)	(0.258)	(0.259)
2018Q4	-1.150***	-1.156***	-1.560***	-2.031***	-1.720***	-1.886***
	(0.244)	(0.270)	(0.275)	(0.286)	(0.299)	(0.302)
2019Q1	-0.192	-0.200	-0.813*	-0.881**	-0.472	-0.738*
	(0.266)	(0.306)	(0.316)	(0.298)	(0.322)	(0.329)
2019Q2	-1.272***	-1.281***	-1.722***	-2.150***	-1.636***	-1.827***
	(0.289)	(0.345)	(0.351)	(0.341)	(0.375)	(0.378)
2019Q3	-2.721***	-2.733***	-3.177***	-2.626***	-2.005***	-2.198***
	(0.320)	(0.391)	(0.396)	(0.372)	(0.416)	(0.419)
2019Q4	-2.016***	-2.030***	-2.474***	-2.811***	-2.090***	-2.282***
	(0.344)	(0.432)	(0.438)	(0.389)	(0.444)	(0.447)
2020Q1	-2.235***	-2.250***	-2.688***	-3.269***	-2.453***	-2.644***
	(0.366)	(0.473)	(0.478)	(0.416)	(0.480)	(0.483)
2020Q2	-8.640***	-8.658***	-9.500***	-11.08***	-10.16***	-10.53***
	(0.423)	(0.544)	(0.556)	(0.483)	(0.556)	(0.564)
2020Q3	-6.181***	-6.200***	-7.040***	-6.295***	-5.266***	-5.640***
	(0.438)	(0.581)	(0.591)	(0.493)	(0.578)	(0.586)
2020Q4	-5.786***	-5.807***	-7.121***	-6.840***	-5.704***	-6.289***
	(0.450)	(0.616)	(0.639)	(0.503)	(0.603)	(0.619)
2021Q1	-4.989***	-5.013***	-5.859***	-5.416***	-4.180***	-4.560***
	(0.476)	(0.663)	(0.673)	(0.529)	(0.639)	(0.646)
2021Q2	-3.966***	-3.992***	-4.865***	-3.975***	-2.635***	-3.024***
	(0.494)	(0.701)	(0.713)	(0.550)	(0.675)	(0.682)
2021Q3	-5.689***	-5.716***	-6.235***	-5.504***	-4.058***	-4.290***
	(0.503)	(0.737)	(0.743)	(0.560)	(0.700)	(0.703)
2021Q4	-5.704***	-5.734***	-6.650***	-5.846***	-4.300***	-4.712***
	(0.526)	(0.782)	(0.794)	(0.585)	(0.739)	(0.747)
Constant	30.33***	30.03***	31.38***	38.82***	54.45***	54.98***
	(0.286)	(6.017)	(6.044)	(0.326)	(4.413)	(4.414)
Observations	163049	163049	163049	148648	148648	148648
R^2	0.017	0.017	0.018	0.020	0.020	0.021

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$